Research Question: "How can Principal Component Analysis (PCA) be applied to uncover key patterns in hospital patient data?"

In [1]:
```
#Importing Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import zscore
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
```

In [2]:
```
#Loading the dataset
medical_clean = pd.read_csv('medical_clean_D212.csv')
```

In [3]:
```
#Information about the dataset
medical_clean.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 50 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   CaseOrder           10000 non-null  int64
 1   Customer_id         10000 non-null  object
 2   Interaction         10000 non-null  object
 3   UID                 10000 non-null  object
 4   City                10000 non-null  object
 5   State               10000 non-null  object
 6   County              10000 non-null  object
 7   Zip                 10000 non-null  int64
 8   Lat                 10000 non-null  float64
 9   Lng                 10000 non-null  float64
 10  Population          10000 non-null  int64
 11  Area                10000 non-null  object
 12  TimeZone            10000 non-null  object
 13  Job                 10000 non-null  object
 14  Children            10000 non-null  int64
 15  Age                 10000 non-null  int64
 16  Income              10000 non-null  float64
 17  Marital             10000 non-null  object
 18  Gender              10000 non-null  object
 19  ReAdmis             10000 non-null  object
 20  VitD_levels         10000 non-null  float64
 21  Doc_visits          10000 non-null  int64
 22  Full_meals_eaten    10000 non-null  int64
 23  vitD_supp           10000 non-null  int64
 24  Soft_drink          10000 non-null  object
 25  Initial_admin       10000 non-null  object
 26  HighBlood           10000 non-null  object
 27  Stroke              10000 non-null  object
 28  Complication_risk   10000 non-null  object
 29  Overweight          10000 non-null  object
 30  Arthritis           10000 non-null  object
 31  Diabetes            10000 non-null  object
 32  Hyperlipidemia      10000 non-null  object
 33  BackPain            10000 non-null  object
 34  Anxiety             10000 non-null  object
 35  Allergic_rhinitis   10000 non-null  object
 36  Reflux_esophagitis  10000 non-null  object
 37  Asthma              10000 non-null  object
 38  Services            10000 non-null  object
 39  Initial_days        10000 non-null  float64
 40  TotalCharge         10000 non-null  float64
 41  Additional_charges  10000 non-null  float64
 42  Item1               10000 non-null  int64
 43  Item2               10000 non-null  int64
 44  Item3               10000 non-null  int64
 45  Item4               10000 non-null  int64
 46  Item5               10000 non-null  int64
 47  Item6               10000 non-null  int64
 48  Item7               10000 non-null  int64
 49  Item8               10000 non-null  int64
dtypes: float64(7), int64(16), object(27)
memory usage: 3.8+ MB
```

In [4]:

```
#Selecting continuous variables
variables = ['Age', 'Income', 'VitD_levels', 'Initial_days', 'TotalCharge', 'Additional_charges'
```

```
df = medical_clean[variables].copy()
```

In [5]:
```
#Checking for duplicate values
print(df.duplicated())
```

```
0       False
1       False
2       False
3       False
4       False
        ...
9995    False
9996    False
9997    False
9998    False
9999    False
Length: 10000, dtype: bool
```

In [6]:
```
#Checking for missing values
print(df.isnull().sum())
```

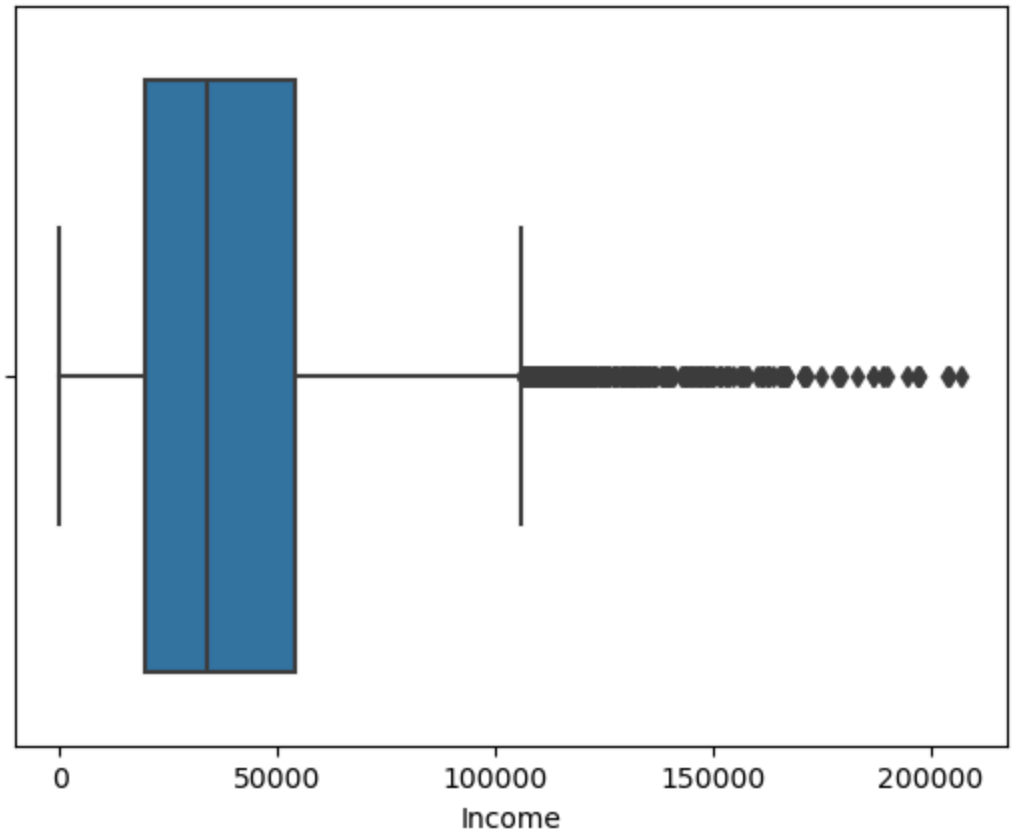
```
Age                   0
Income                0
VitD_levels           0
Initial_days          0
TotalCharge           0
Additional_charges    0
dtype: int64
```

In [7]:
```
#Identifying outliers based on Z-scores
z_scores = np.abs(zscore(df[variables]))
z_score_outliers = (z_scores > 3).sum(axis=0)
print("Outliers based on Z-score method:")
print(z_score_outliers)
```

```
Outliers based on Z-score method:
Age                     0
Income                143
VitD_levels            24
Initial_days            0
TotalCharge             0
Additional_charges      0
dtype: int64
```

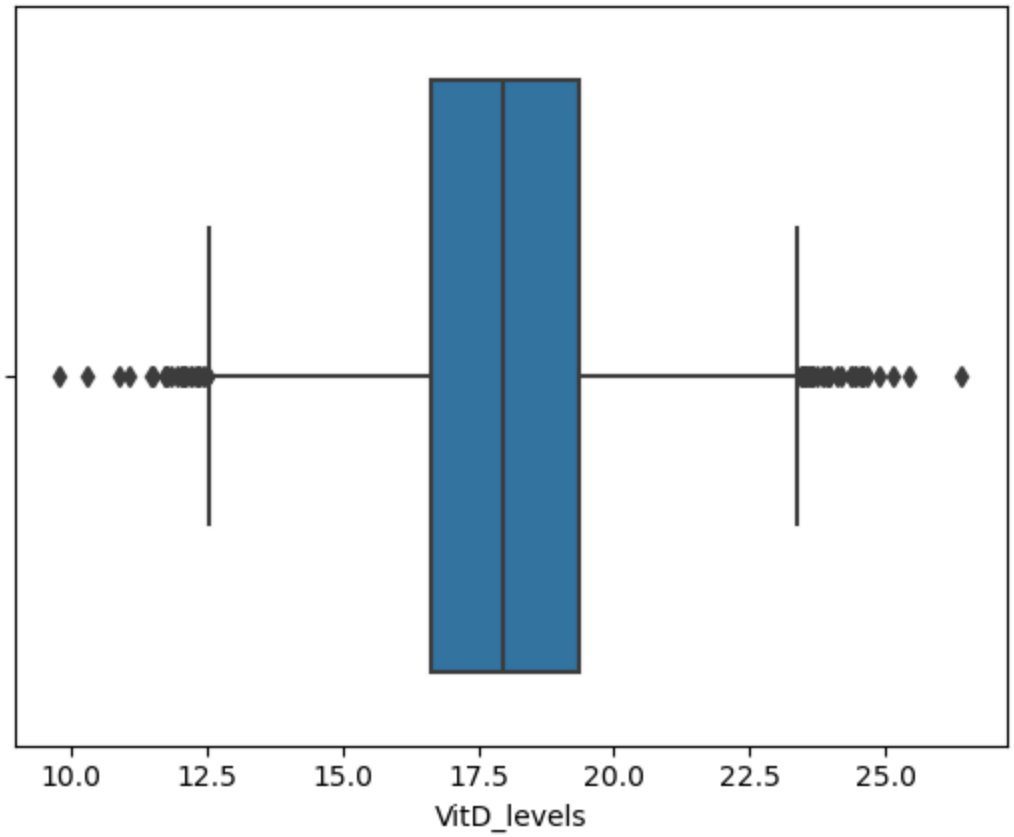
In [8]:
```
#Checking for outliers for Income
sns.boxplot(x = 'Income', data = df)
```

Out[8]:
```
<Axes: xlabel='Income'>
```



In [9]:
```
#Checking for outliers for VitD_levels
sns.boxplot(x = 'VitD_levels', data = df)
```

Out[9]: <Axes: xlabel='VitD_levels'>



In [10]:
```
#Separate features from label
X = df
```

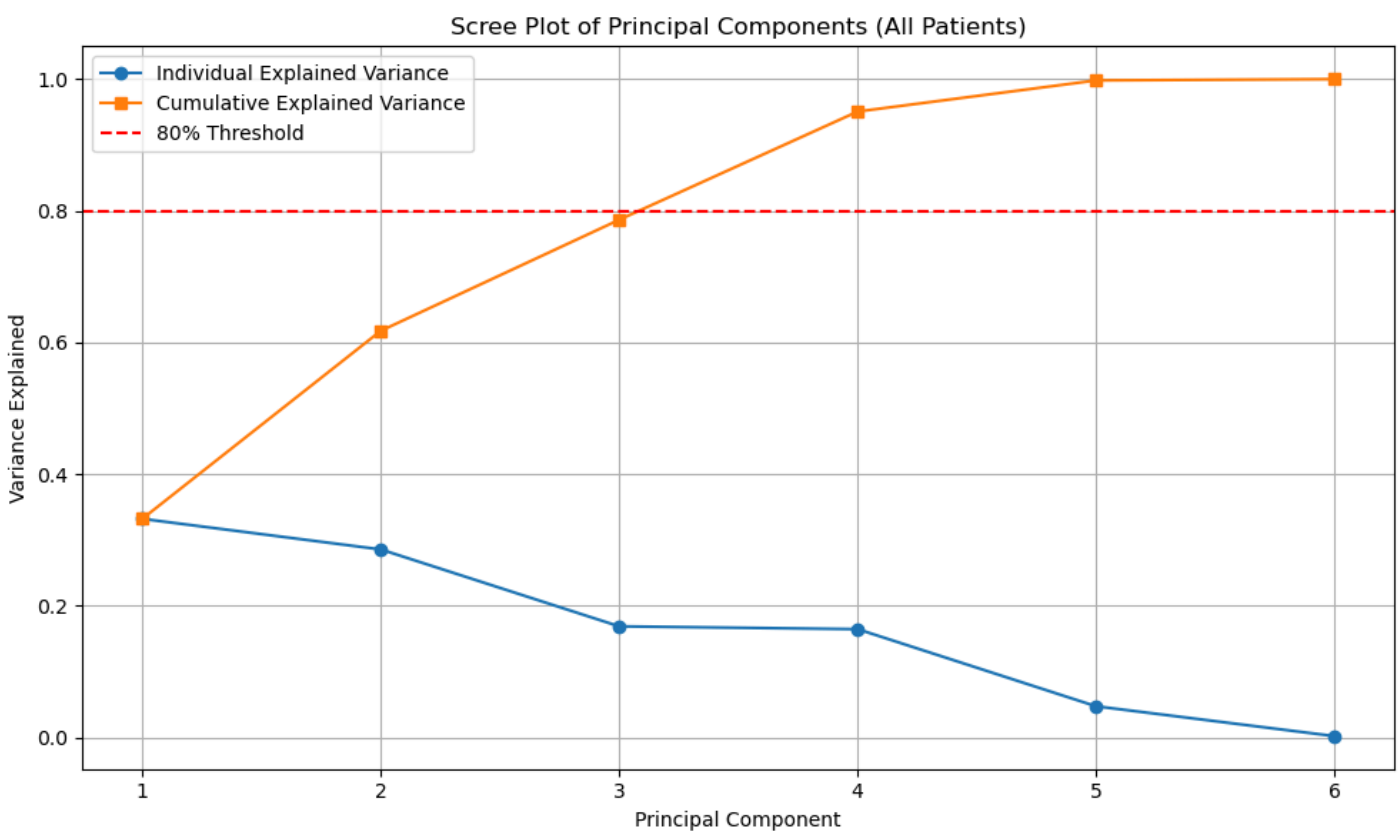
```
y = medical_clean['ReAdmis']
```

In [11]:
```
#Feature scaling
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

In [12]:
```
#PCA
pca = PCA()
X_pca = pca.fit_transform(X_scaled)
```

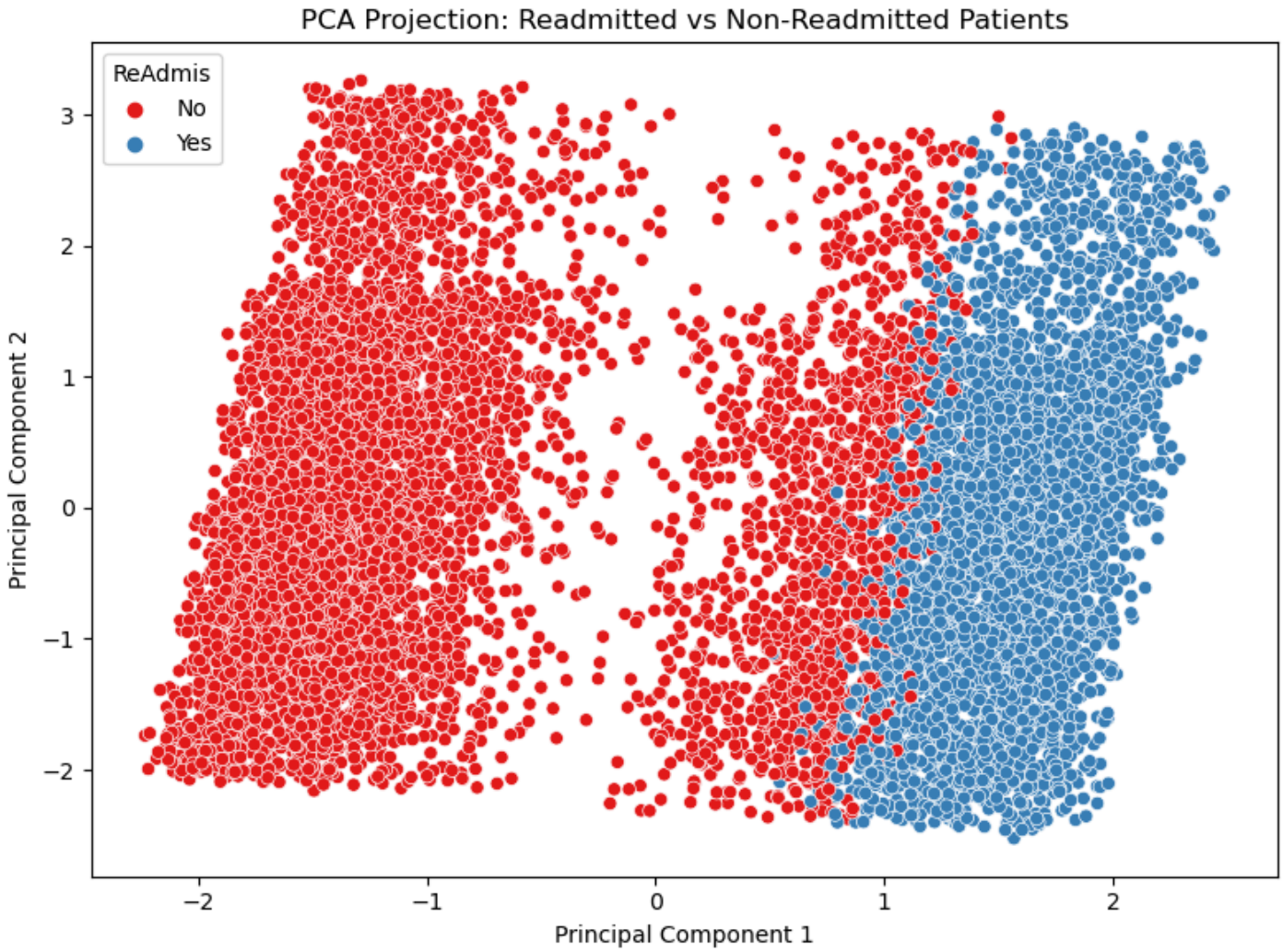
In [13]:
```
#Explained and cumulative variance
explained_variance = pca.explained_variance_ratio_
cumulative_variance = np.cumsum(explained_variance)
```

In [14]:
```
#Scree plot
plt.figure(figsize=(10, 6))
plt.plot(range(1, len(explained_variance) + 1), explained_variance, marker='o', label='Individua
plt.plot(range(1, len(cumulative_variance) + 1), cumulative_variance, marker='s', label='Cumulat
plt.axhline(y=0.8, color='r', linestyle='--', label='80% Threshold')

plt.title("Scree Plot of Principal Components (All Patients)")
plt.xlabel("Principal Component")
plt.ylabel("Variance Explained")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```



In [15]:
```
#Scatterplot colored by readmisiion
pca_df = pd.DataFrame(X_pca, columns=[f'PC{i+1}' for i in range(X.shape[1])])
pca_df['ReAdmis'] = y.values

plt.figure(figsize=(8, 6))
```

```
sns.scatterplot(data=pca_df, x='PC1', y='PC2', hue='ReAdmis', palette='Set1')
plt.title('PCA Projection: Readmitted vs Non-Readmitted Patients')
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.legend(title="ReAdmis")
plt.tight_layout()
plt.show()
```



In [16]:
```
#Variance exp
explained_variance = pca.explained_variance_ratio_
cumulative_variance = np.cumsum(explained_variance)
```

In [17]:
```
#Varience summary
print("Explained Variance (First 4 PCs):")
for i in range(4):
    print(f"PC{i+1}: {explained_variance[i]:.4f} (Cumulative: {cumulative_variance[i]:.4f})")
```

```
Explained Variance (First 4 PCs):
PC1: 0.3320 (Cumulative: 0.3320)
PC2: 0.2856 (Cumulative: 0.6176)
PC3: 0.1687 (Cumulative: 0.7863)
PC4: 0.1644 (Cumulative: 0.9508)
```

In [18]:
```
#Get loadings
loadings = pd.DataFrame(
    pca.components_.T,
    columns=[f'PC{i+1}' for i in range(len(variables))],
    index=variables
```

```
)

print("\nTop Variables Contributing to PC1–PC4:")
print(loadings.iloc[:, :4].sort_values(by='PC1', ascending=False))
```

```
Top Variables Contributing to PC1–PC4:
                        PC1       PC2       PC3       PC4
TotalCharge        0.702270 -0.080918 -0.005929  0.011405
Initial_days       0.701146 -0.091464 -0.008280  0.011129
Additional_charges 0.086047  0.701555 -0.022194  0.001522
Age                0.085851  0.701556 -0.017940  0.001094
VitD_levels       -0.001680  0.019269  0.712533  0.701368
Income            -0.020851 -0.018816 -0.700984  0.712619
```

In [19]:
```
#Total variance captured by the first 6 components
total_variance_4 = cumulative_variance[3] * 100
print(f"Total variance explained by first 4 components: {total_variance_4:.2f}%")
```

```
Total variance explained by first 4 components: 95.08%
```

In [20]:
```
# Number of components that explain at least 80% variance
n_components_80 = np.argmax(cumulative_variance >= 0.80) + 1
print(f"Number of components explaining >= 80% variance: {n_components_80}")
```

```
Number of components explaining >= 80% variance: 4
```

In [21]:
```
#Convert scaled numpy array to a dataframe
scaled_df = pd.DataFrame(X_scaled, columns=variables)
```

In [22]:
```
#Variance summary dataframe
variance_df = pd.DataFrame({
    'Explained Variance': explained_variance,
    'Cumulative Variance': cumulative_variance
})
```

In [23]:
```
#Export data
scaled_data_path = 'output/scaled_medical_data_PCA_D212_II.csv'
scaled_df.to_csv(scaled_data_path, index=False)
new_data_path = 'output/new_medical_data_PCA_D212_II.csv'
df.to_csv(new_data_path, index=False)
variance_path =new_data_path = 'output/variance_summary_PCA_D212_II.csv'
variance_df.to_csv(variance_path, index=False)
loadings_path = 'output/loadings_PCA_D212_II.csv'
loadings.to_csv(loadings_path)
```

In [ ]: